

```

    return board;
}

function solve(board, maxMoves) {
    // Attempt to solve the puzzle in `board` in at most `maxMoves`
    // moves. Returns true if solved, otherwise false.
    document.write("Attempting to solve in at most " + maxMoves + " moves...<br /
>");
    let solutionMoves = []; // A list of UP, DOWN, LEFT, RIGHT values.
    let solved = attemptMove(board, solutionMoves, maxMoves, null);

    if (solved) {
        displayBoard(board);
        for (let move of solutionMoves) {
            document.write("Move " + move + "<br />");
            makeMove(board, move);
            document.write("<br />"); // Print a newline.
            displayBoard(board);
            document.write("<br />"); // Print a newline.
        }
        document.write("Solved in " + solutionMoves.length + " moves:<br />");
        document.write(solutionMoves.join(", ") + "<br />");
        return true; // Puzzle was solved.
    } else {
        return false; // Unable to solve in maxMoves moves.
    }
}

```

```

function attemptMove(board, movesMade, movesRemaining, prevMove) {
    // A recursive function that attempts all possible moves on `board`
    // until it finds a solution or reaches the `maxMoves` limit.
    // Returns true if a solution was found, in which case `movesMade`
    // contains the series of moves to solve the puzzle. Returns false
    // if `movesRemaining` is less than 0.

    if (movesRemaining < 0) {
        // BASE CASE - Ran out of moves.
        return false;
    }

    if (JSON.stringify(board) == SOLVED_BOARD) {
        // BASE CASE - Solved the puzzle.
        return true;
    }
}

```